ML-Driven Localized control system for simplified Net Metering and voltage Regulation in urban PV-integrated Distribution networks in Bangladesh.

Md. Sayed Parvej, Md. Ratul Hasan

Dept. of Electrical and Electronic Engineering, Jashore University of science and Technology

Introduction

- Rapid rooftop PV adoption in Bangladesh's urban feeders creates voltage instability and bidirectional energy flow challenges.
- Net metering requires accurate, real-time energy tracking and settlement, which legacy systems struggle to provide.
- This project develops a **machine learning-based localized control system** to:
  - •Forecast PV generation and load demand.
  - •Regulate voltage autonomously using smart inverter and feeder controls.
  - •Simplify net metering through accurate bidirectional energy accounting.
- Traditional control methods (Volt/VAR, OLTC tap changes) are insufficient under high PV variability.

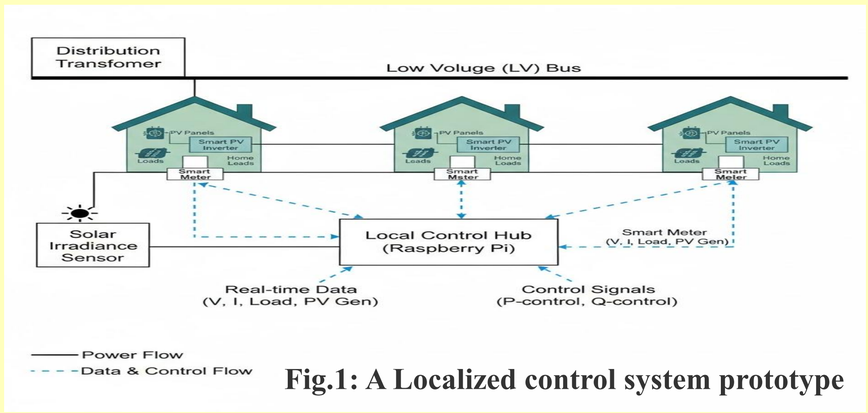


Fig.1: A Localized control system prototype

Motivation

- **For Users:** Simplify net-metering and add battery backup, making solar reliable and profitable.
- **For Utilities:** Enhance the stability and capacity of the Low Voltage (LV) distribution network using localized, predictive control.
- **For Bangladesh:** Safely integrate more renewable energy to reduce national dependency on fossil fuels.

Block Diagram

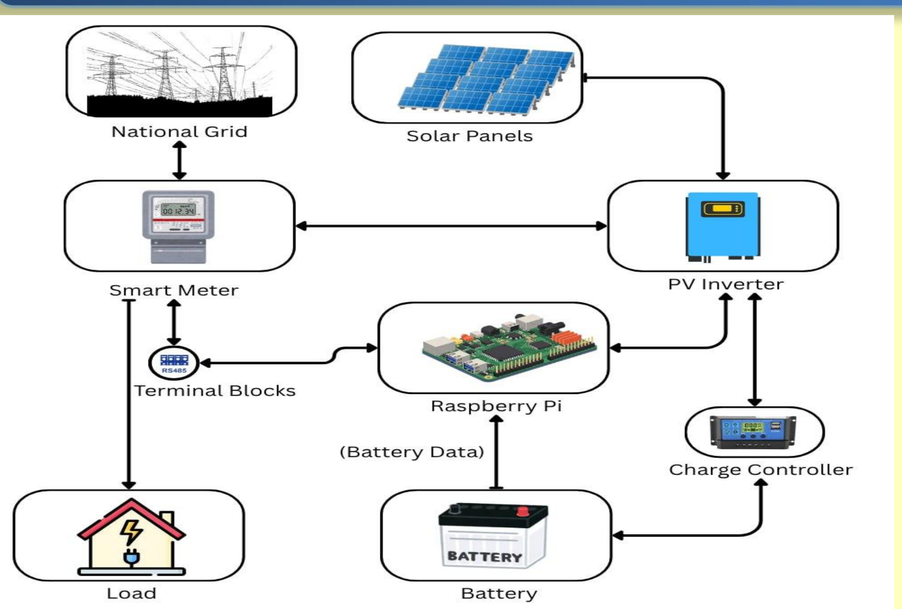
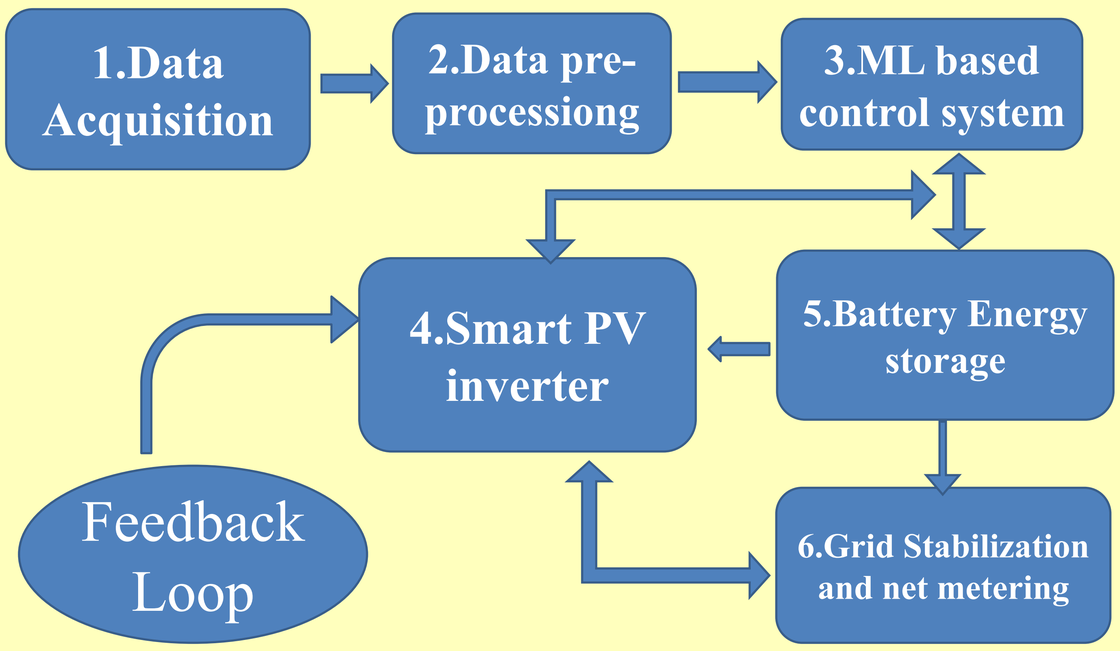


Fig. 2: System schematic diagram

Methodology



Expected Results

The ML controller significantly improves voltage stability, reduces tap changes, minimizes feeder losses, and enhances net metering accuracy compared to traditional control methods. These results demonstrate the effectiveness of machine learning in optimizing smart grid operations under high rooftop PV penetration.

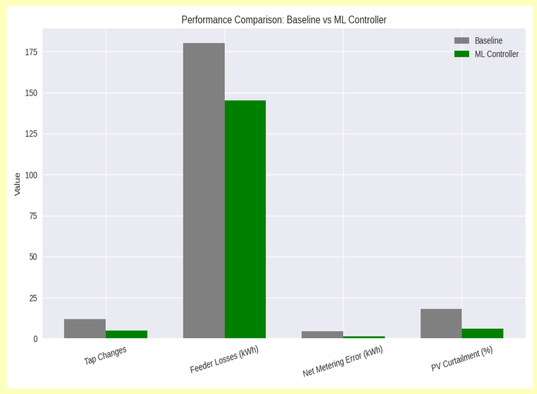


Fig.3: Improved performance parameter of the proposed model

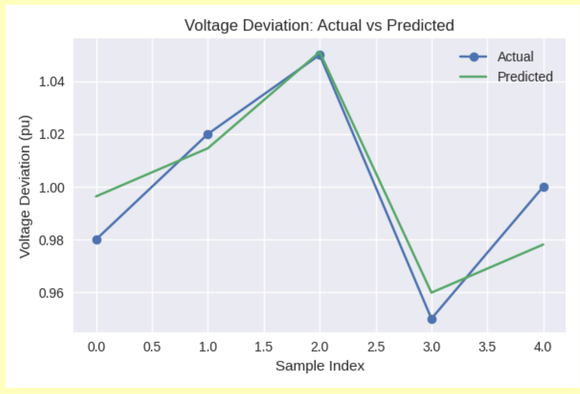


Fig.4:Proposed ML model predicts voltage deviation based on PV generation and load demand.

Model proposed: Reinforcement Learning with LSTM forecasting support.

Outcomes & Impacts:

- ❖Voltage regulation improved with fewer violations in PV-integrated urban feeders.
- ❖Feeder losses reduced compared to conventional control method. OLTC tap changes minimized, lowering mechanical wear.
- ❖Net metering accuracy enhanced with reliable bidirectional energy tracking.
- ❖Cost Advantage: Low-cost solution using open-source tools and existing grid hardware — no expensive upgrades required

Cost Analysis

Name of Components	Purpose	Estimated Cost (BDT)
Raspberry Pi 4 (4GB)	Main Control Hub processor	৳11,000
4G LTE Modem/Router	Communication (Data to/from devices)	৳2,500
Weatherproof Enclosure (IP65)	Protects electronics outdoors	৳1,500
Raspberry Pi Power Supply (5V, 3A)	Power for the hub	৳800
Solar Irradiance Sensor (I2C)	ML Model Input (As data can come from inverter)	৳400
USB to RS485 Adapter	Communication with Modbus devices	৳350
32GB MicroSD Card (Class 10)	OS and Data Storage	৳400
Misc. (Wires, Connectors, Mounts)	Assembly	৳300
Total costs		≈17,250

Conclusions

In conclusion, the proposed system demonstrates the feasibility of implementing a cost-effective, ML-driven localized control hub using accessible IoT technology. It provides valuable support for real-time voltage regulation, simplified net metering, and resilient urban power distribution, effectively encouraging sustainable solar adoption in Bangladesh.

References

[1] M. Hossain, et al., *Autonomous Voltage Regulation for Smart Distribution Networks with High PV Penetration*, IEEE Transactions on Smart Grid, 2024.

[2] S. Kumar, et al., *Smart Control Strategies for Voltage Regulation in Grid-Tied PV Systems*, IEEE Access, 2023.

[3] R. Sharma, et al., *Advanced Control Strategies for Resilient Voltage and Frequency Regulation in Power Systems*, Int. Res. Open J. of Electrical Engineering Applications, 2022.

Dept. of Electrical and Electronic Engineering
Jashore University of Science & Technology,
Jashore-7408, Bangladesh